

# C é s a r   C a r v a j a l   d e   l a   H a z a

Mathematician, Statistician and Physicist

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## Qualification Summary

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Mathematics, Statistics and Physics graduate. Strong foundation in **quantum mechanics, theoretical physics, and statistical methods**. Passionate about **problem-solving, coding, and data analysis**, with practical experience in Python, R, and mathematical modeling. Interested in **quantum computing, artificial intelligence and the space**. Committed to contributing to scientific progress and technological advancements.

## Education

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### **Bachelor's Degree in Physics, 240 ECTS (2021 - 2025) (GPA: 8.1/10)**

National University of Distance Education (UNED), Spain

- ♦ Strong background in **classical and quantum physics, optics and dynamical systems**.
- ♦ Fields of interest: Quantum Physics, General Relativity, Theoretical Physics, Nuclear Physics, Optics.
- ♦ Highlights:
  - ♦ I particularly enjoyed courses involving mathematical and conceptual problem-solving, especially **Quantum Physics** and **Dynamical Systems**.
  - ♦ I developed a solid understanding of how physical models are built and how mathematics shapes physical intuition.
  - ♦ Honors in Classical Mechanics, Vibrations and Waves, Quantum Physics II and Dynamical Systems.
- ♦ **Final Grade Project**: Quantum Mechanics and the study of **Entanglement Entropy and Geometry**. Implemented simulations in **Python** to analyze entanglement in quantum systems.

### **Bachelor's Degree in Mathematics, 240 ECTS (2016 - 2021) (GPA: 8.0/10)**

University of Seville, Spain

- ♦ Strong foundation in **calculus, algebra and differential equations**.
- ♦ Fields of interest: Computing (IT), Basic Algebra, Ordinary Differential Equations, Topology, Probability Theory, Algebraic Structures, Commutative Algebra and Algebraic Geometry, Coding Theory and Cryptography, Partial Differential Equations, Computer Science.
- ♦ Highlights:
  - ♦ I learned some programming languages such as **Haskell, Python, Maxima** and **MATLAB**. I built a strong and solid foundation about coding and programming, in general.
  - ♦ I really enjoyed Computing. It was one of my favourite courses I took during this Degree. It consisted mainly in solving mathematical problems through coding.
  - ♦ My favourite field is Algebra. I love numbers and the structures they may form. I would love to keep learning more about it.

## Bachelor's Degree in Statistics, 240 ECTS (2016 - 2021) (GPA: 8.5/10)

University of Seville, Spain

- ♦ Expertise in **probability, statistical inference** and **data analysis**.
- ♦ Fields of interest: Statistical Inference, Databases, Survey Design, Operations Research, Multivariate Analysis, Computational Statistics, Decision Theory, Artificial Intelligence and Statistics, Algebraic Methods in Probability and Statistics.
- ♦ Highlights:
  - ♦ I developed many projects through **R**, and also with **Excel**.
  - ♦ I really enjoyed the whole experience of data treatment and analysis.
  - ♦ I got some experience with a project during the "Survey Design" course, where I got to work with other students and professor. I also passed this course with honours.
  - ♦ I was introduced to **Artificial Intelligence** during the final year, and also worked with **Machine Learning** algorithms throughout the whole program.

## Work & Experience

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### GNSS Trainee | GNSS Academy (January 2025 to July 2025)

- ♦ 6-month intensive program focused on **Global Navigation Satellite System** (GNSS).
- ♦ Developed a **GNSS+IMU sensor fusion** simulator using **Python** and **Linux**.
- ♦ Simulated satellite orbits and GNSS measurements with realistic errors.
- ♦ Gained solid understanding of GNSS architecture and satellite-based positioning.
- ♦ Hands-on experience with **gLAB** for analyzing GNSS observables and signals.
- ♦ Introduced to advanced positioning techniques including **RTK**, **PPP**, and **DGNSS**.
- ♦ About the **project**:
  - ♦ Developed a full-featured simulation environment to test GNSS/IMU sensor fusion algorithms for integrated navigation.
  - ♦ Simulated GNSS measurements using precise SP3 orbit propagation and realistic error models (ionospheric, tropospheric, clock, multi path).
  - ♦ Applied Antenna Phase Offset (APO) and Sagnac effect corrections to correct the dynamic of the satellite.
  - ♦ Applied mechanization equations in ECEF frame to derive position, velocity and attitude from simulated IMU data.
  - ♦ Performed coordinate transformations between different coordinate systems to ensure accurate motion tracking.
  - ♦ Implemented a Least-Squares Estimation module for GNSS-only positioning.

### QA Automation Engineer | Deloitte (April 2022 to November 2022)

- ♦ Developed automated tests using **Selenium** and **Java**, decreasing manual testing time and improving the quality of many tests.
- ♦ Helped implementing an automation framework using **Selenium**, **Cucumber** and **Jenkins**, improving the testing efficiency and coverage.
- ♦ Executed test scripts and documented results, ensuring a thorough testing process.
- ♦ Created detailed bug reports and collaborated with developers to resolve issues.
- ♦ Identified, prioritized and resolved defects using **Jira**.
- ♦ Used **Git** to have synchronous code from the whole team.
- ♦ Collaborated with cross-functional teams in an international environment.

### Personal Teacher (Several years)

- ♦ Tutored students in **Mathematics, Physics, Chemistry and Technical Drawing** improving their academic performance.
- ♦ Created a positive environment for the students to feel free and discuss relevant and important ideas for their subjects.
- ♦ Focus on clarity, logic and problem-solving.

## Research and Academic/Personal Projects

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### Learning Linux, C/C++ and Bash (ongoing)

- ♦ Started learning **Linux** basics, **Bash** scripting, and **C/C++** programming.
- ♦ Completed small exercises to practice scripting, file handling, and simple programs.
- ♦ Gaining foundational skills in automation and scripting using Bash.

### Entanglement Entropy in Quantum Systems (2024 - 2025)

- ♦ I have studied entanglement entropy and its dependence on system geometry in order to test the limits of the Area Law.
- ♦ I have implemented simulations and numerical calculations using **Python**.
- ♦ **Code available on GitHub:** <https://github.com/cesardelahaza>.

### Survey Design and Statistical Analysis (Course 2019/20)

- ♦ Participated in the **2019/20 User Satisfaction Study** of the IT Service at the University of Seville.
- ♦ Contributed to survey design, data collection, data analysis, and final report preparation.
- ♦ Applied statistical methods in R and Excel for data analysis.

### Problem-Solving and Coding Challenges (2016 - 2018)

- ♦ Worked on coding problems on **Exercitium** and **Kattis** to develop problem-solving and algorithmic skills.

## Technical Skills

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- ♦ **Programming Languages:** Python, R, Haskell, Matlab, SQL (basic), C/C++ (basic), Bash (basic).
- ♦ **Tools & Platforms:** Linux, Git, VSCode, Jupyter, LaTeX.
- ♦ **Data Analysis & Modeling:** Applied statistics, mathematical modeling, machine learning, optimization.
- ♦ **Applied Mathematics:** Algebra, optimization, differential equations, and cryptography.

## Soft Skills

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|------------------------|-----------------------|
| ♦ Critical Thinking    | ♦ Critical Thinking   |
| ♦ Analytical Skills    | ♦ Attention to Detail |
| ♦ Creative Thinking    | ♦ Adaptability        |
| ♦ Interpersonal Skills | ♦ Research Skills     |

## Extracurricular Activities & Volunteering

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### LGTB+ Advocacy & Awareness (2017-2019)

- ♦ Active member of **Colectivo LGTB+ Pedro Zero** at the Faculty of Mathematics, University of Seville.
- ♦ Participated in **awareness campaigns and outreach activities** in topics related to the LGBT+ community.

### Animal Welfare Volunteering (Teenage years)

- ♦ Provided weekly care for rescued dogs, ensuring **clean living spaces, socialization, and exercise**.
- ♦ Created **adoption advertisements** and participated in awareness campaigns about responsible pet ownership.

## Languages

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**Spanish** (Native)  
**English** (C1 - Proficient)  
**French** (Basic, in process)

## More About Me

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- ♦ **Interests:** Passionate about **Quantum Physics, Artificial Intelligence, and Data Visualization.**
- ♦ **Hobbies:** Walking my dog, doing **Crossfit**, and writing. Also passionate about **learning new languages.**
- ♦ **Beyond Work:** Always open to **new challenges**, whether in a book, a coding problem, or any sport activity.